3. Effect of temperature

When temperature is raised chemisorption first increases and then decreases. whereas physisorption decreases with increase in temperature.

4. Effect of pressure:

chemical adsorption is fast with increase in pressure, it can not alter the amount of adsorption. In Physisorption the extent of adsorption increases with increase in pressure.

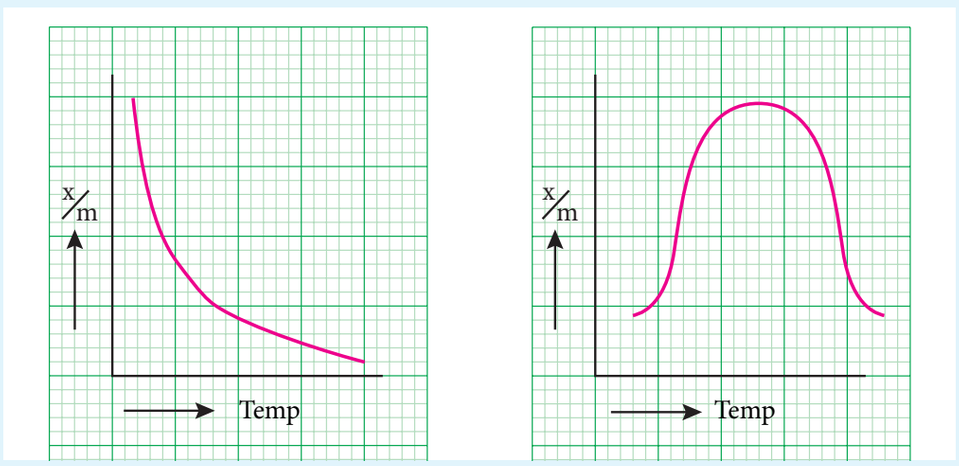
10.1.3 Adsorption isotherms and isobars.

Adsorption isotherms represents the variation of adsorption at constant temperature.

When amount of adsorption is plotted versus temperature at constant pressure it is called adsorption isobar.

Adsorption isobars of physisorption and chemisorption are different as represented in the graphs.

Figure 10.1 (a) Physical Adsorption **Figure 10.1 (b) Chemical Adsorption**



x is the amount of adsorbate adsorbed on 'm' g of adsorbent.

In physical adsorption, $\frac{x}{m}$ decreases with increase in Temprature, But in chemical adsorption, $\frac{x}{m}$ increases with rise in temperature and then decreases. The increase illustrates the requirement of activation of the surface for adsorption is due to fact that formation of activated complex requires certain energy.

The decrease at high temperature is due to desorption, as the kinetic energy of the adsorbate increases.

10.1.3.1 Adsorption isotherms

Adsorption isotherm can be studied quantitatively. A plot between the amount of adsorbate adsorbed and pressure (or concentration of adsorbate) at constant temperature is called adsorption isotherms.

In order to explain these isotherms various equations were suggested as follows:

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